# Unit 5

# Equilibrium Effects

## Objectives of the Study:

After studying this unit, the student will be able

- to know how the equilibrium effects of Government interventions

## Equilibrium Effects of Government Interventions

Government interventions can significantly impact market equilibrium by altering supply and demand dynamics. Policies that shift supply curves often involve taxes, subsidies, or regulations that affect production costs. These interventions can lead to higher prices, reduced quantities supplied, or both. Policies that cause demand to differ from supply can create surpluses or shortages, depending on whether the intervention affects supply or demand more. Importantly, supply and demand need not always equal each other in a market due to various factors like government interventions, external shocks, or changes in consumer preferences.

## Policies That Shift Supply Curves

Governments employ a variety of policies that shift supply curves. Two common policies are licensing laws and quotas.

### a) Licensing Laws

A government licensing law limits the number of firms that may sell goods in a market. For example, many local governments around the world limit the number of taxicabs. Governments use zoning laws to limit the number of bars, bookstores, hotel chains, as well as firms in many other markets. In developed countries, licenses are distributed to early entrants or exams are used to determine who is licensed. In developing countries, licenses often go to relatives of government officials or to whomever offers those officials the largest bribe. determine who is licensed. In developing countries, licenses often go to relatives of government officials or to whomever offers those officials the largest bribe.

## Occupational Licensing

Occupational licensing is the application of Licensing laws. Many occupations are licensed in the United States. In those occupations, working without a license is illegal. More than 800 occupations are licensed at the local, state, or federal level, including animal trainers, dietitians and nutritionists, doctors, electricians, embalmers, funeral directors, hair dressers, librarians, nurses, psychologists, real estate brokers, respiratory therapists, salespeople, teachers, and tree trimmers.

During the early 1950s, fewer than 5 percent of U.S. workers were in occupations covered by licensing laws at the state level. Since then, the share of licensed workers has grown, reaching nearly 18 percent by the 1980s, at least 20 percent in 2000, and 29 percent in 2008. Licensing is more common in occupations that require extensive education: More than 40 percent of workers with post-college education are required to have a license compared to only 15 percent of those with less than a high school education.

In some occupations to get licensed one must pass a test, which is frequently designed by licensed members of the occupation. By making the exam difficult, current workers can limit entry. For example, only 42 percent of people taking the California State Bar Examination in 2011 and 2012 passed it, although all of them had law degrees.

To the degree that testing is objective, licensing may raise the average quality of the workforce. However, its primary effect is to restrict the number of workers in an occupation. To analyze the effects of licensing, where the wage is on the vertical axis and the number of workers per year is on the horizontal axis. Licensing shifts the occupational supply curve to the left, reducing the equilibrium quantity of workers and raising the wage. Kleiner and Krueger (2013) found that licensing raises occupational wages by 18 percent.

## Quotas

Quotas typically limit the amount of a good that can be sold (rather than the number of firms that sell it). Quotas are commonly used to limit imports. Quotas on imports affect the supply curve. We illustrate the effect of quotas on market equilibrium.

The Japanese government's ban (the quota was set to zero) on rice imports raised the price of rice in Japan substantially. Figure 5.1 shows the Japanese demand curve for rice, D, and the total supply curve without a ban, S. The intersection of S and D determines the equilibrium, $e_1$, if rice imports are allowed

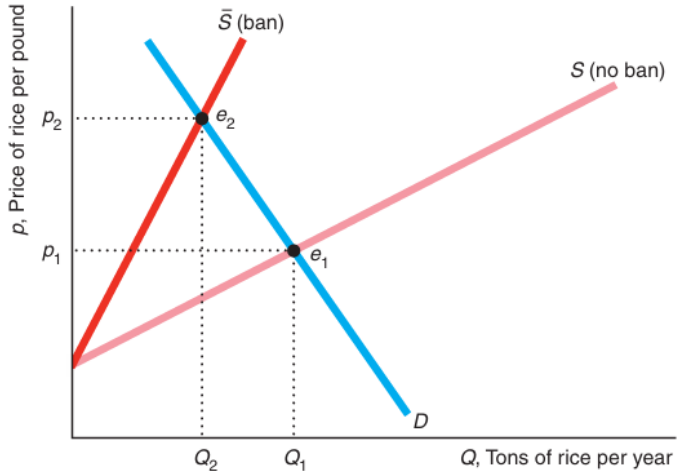



**Figure 5.1 A Ban on Rice Imports Raises the Price in Japan**

A ban on rice imports shifts the total supply of rice in Japan without a ban, S, to $\bar{S}$, which equals the domestic supply alone. As a result, the equilibrium changes from $e_1$ to $e_2$. The ban causes the price to rise from $p_1$ to $p_2$ and the equilibrium quantity to fall to $Q_1$ from $Q_2$.

What is the effect of a ban on foreign rice on Japanese supply and demand? The ban has no effect on demand if Japanese consumers do not care whether they eat domestic or foreign rice. The ban causes the total supply curve to rotate toward the origin from S (total supply is the horizontal sum of domestic and foreign supply) to S (total supply equals the domestic supply).

The intersection of S and D determines the new equilibrium, $e_2$, which lies above and to the left of $e_1$. The ban caused a shift of the supply curve and a movement along the demand curve. It led to a fall in the equilibrium quantity from $Q_1$ to $Q_2$ and a rise in the equilibrium price from $p_1$ to $p_2$. Because of the Japanese nearly total ban on imported rice, the price of rice in Japan was 10.5 times higher than the price in the rest of the world in 2001, but is only about 50 percent higher today.

A quota of Q may have a similar effect to an outright ban; however, a quota may have no effect on the equilibrium if the quota is set so high that it does not limit imports.

**Policies That Cause Demand to Differ from Supply**

Some government policies do more than merely shift the supply or demand curve. For example, governments may control prices directly, a policy that leads to either excess supply or excess demand if the price the government sets differs from the equilibrium price. We illustrate this result with two types of price control programs: price ceilings and price floors. When the government sets a price ceiling at p, the price at which goods are sold may be no higher than p. When the government sets a price floor at p, the price at which goods are sold may not fall below p.

**a) Price Ceilings**

Price ceilings have no effect if they are set above the equilibrium price that would be observed in the absence of the price controls. If the government says that firms may charge no more than $\bar{p}$ = \$5 per gallon of gas and firms are actually charging p = \$1, the government's price control policy is irrelevant. However, if the equilibrium price, p, was above the price ceiling $\bar{p}$, the price actually observed in the market would be the price ceiling.

The United States used price controls during both world wars, the Korean War, and in 1971–1973 during the Nixon administration, among other times. The U.S. experience with gasoline illustrates the effects of price controls. In the 1970s, the Organization of Petroleum Exporting Countries (OPEC) reduced supplies of oil (which is converted into gasoline) to Western countries. As a result, the total supply curve for gasoline in the United States - the horizontal sum of domestic and OPEC supply curves - shifted to the left from $S^1$ to $S^2$ in Figure 5.2. Because of this shift, the equilibrium price of gasoline would have risen substantially, from $p_1$ to $p_2$. In an attempt to protect consumers by keeping gasoline prices from rising, the U.S. government set price ceilings on gasoline in 1973 and 1979.

The government told gas stations that they could charge no more than $\bar{p} = p_1$. Figure 5.2 shows the price ceiling as a solid horizontal line extending from the price axis at p. The price control is binding because $p_2 > \bar{p}$. The observed price is the price ceiling. At $\bar{p}$, consumers want to buy $Q_d = Q_1$ gallons of gasoline, which is the equilibrium quantity they bought before OPEC acted.

However, firms supply only $Q_s$ gallons, which is determined by the intersection of the price control line with $S^2$. As a result of the binding price control, excess demand is $Q_d$- $Q_s$.

Were it not for the price controls, market forces would drive up the market price to $p_2$, where the excess demand would be eliminated. The government price ceiling prevents this adjustment from occurring. As a result, an enforced price ceiling causes shortage a persistent excess demand a shortage: a persistent excess demand.

At the time of the controls, some government officials argued that the shortages were caused by OPEC's cutting off its supply of oil to the United States, but that's not true. Without the price controls, the new equilibrium would be $e_2$. In this equilibrium, the price, $p_2$, is much higher than before, $p_1$; however, no shortage results. Moreover, without controls, the quantity sold, $Q_2$, is greater than the quantity sold under the control program, $Q_s$.

With a binding price ceiling, the supply-and-demand model predicts an equilibrium with a shortage. In this equilibrium, the quantity demanded does not equal the quantity supplied. The reason that we call this situation an equilibrium, even though a shortage exists, is that no consumers or firms want to act differently, given the law. Without the price controls, consumers facing a shortage would try to get more output by offering to pay more, or firms would raise prices. With effective government price controls, they know that they can't drive up the price, so they live with the shortage.

Then some lucky consumers get to buy Qs units at the low price of p. Other potential customers are disappointed: They would like to buy at that price, but they cannot find anyone willing to sell gas to them.

What determines which consumers are lucky enough to find goods to buy at the low price when a price control is imposed? With enforced price controls, sellers use criteria other than price to allocate the scarce commodity. Firms may supply their friends, long-term customers, or people of a certain race, gender, age, or religion. They may sell their goods on a first-come, first-served basis. Or they may limit every -one to only a few gallons.

Another possibility is for firms and customers to evade the price controls. A consumer could go to a gas station owner and say, "Let's not tell anyone, but I'll pay you twice the price the government sets if you'll sell me as much gas as I want." If enough customers and gas station owners behaved that way, no shortage would occur. A study of 92 major U.S. cities during the 1973 gasoline price controls found no gasoline lines in 52 of them. However, in cities such as

Chicago, Hartford, New York, Portland, and Tucson, potential customers waited in line at the pump for an hour or more. Deacon and Sonstelie (1989) calculated that for every dollar consumer saved during the 1980 gasoline price controls, they lost $1.16 in waiting time and other factors.

This experience dissuaded most U.S. jurisdictions from imposing gasoline price controls, even when gasoline prices spiked following Hurricane Katrina in the summer of 2008. The one exception was Hawaii, which imposed price controls on the wholesale price of gasoline starting in September 2005, but suspended the controls indefinitely in early 2006 due to the public's unhappiness with the law. However, many other countries have imposed price controls on gasoline more recently, such as Argentina in 2013.

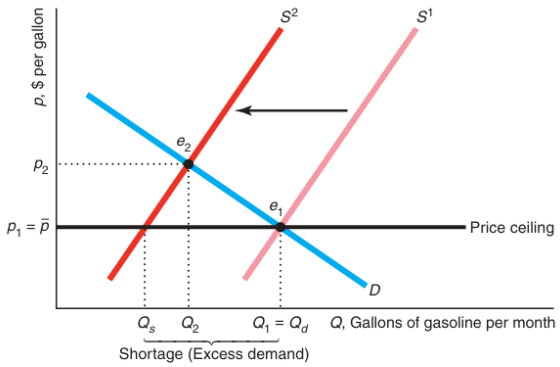



**Figure 5.2 Price Ceiling on Gasoline**

After a shock, the supply curve shifts from $S^1$ to $S^2$. Under the government's price control program, gasoline stations may not charge a price above the price ceiling, $\bar{p} = p_1$. At that price, produc ers are willing to supply only $Q_s$, which is less than the amount $Q_1 = Q_d$ that consumers want to buy. The result is excessive demand, or a shortage of $Q_d - Q_s$

### b) Price Floors

Governments also commonly use price floors. One of the most important examples of a price floor is a minimum wage in a labor market. A minimum wage law forbids employers from paying less than the minimum wage, w. Minimum wage laws date from 1894 in New Zealand, 1909 in the United Kingdom, and 1912 in Massachusetts. The Fair Labor Standards Act of 1938 set a federal U.S. minimum wage of 25¢ per hour (¢ stands for cent which is quarter of a dollar). The U.S. federal minimum hourly wage rose to $7.25 in 2009 and remained at that level through mid-2013, but 19 states have higher state minimum wages. The minimum wage

in Canada differs across provinces, ranging from C$9.50 to C$11.00 (where C$ stands for Canadian dollars, which roughly equal U.S. dollars) in 2013. The U.K. minimum wage for adults is £6.31 ($9.81) as of October 2013. If the minimum wage is binding—that is, if it exceeds the equilibrium wage, w* - it creates unemployment: a persistent excess supply of labor. The original 1938 U.S. minimum wage law caused massive unemployment in Puerto Rico.

**Why Supply Need Not Equal Demand**

Supply and demand are theoretical concepts representing the ideal equilibrium point where quantity supplied and quantity demanded are equal. However, various factors can cause actual market conditions to deviate from this equilibrium. These factors include:

i. **Market disruptions:**

Events like natural disasters or economic crises can temporarily disrupt supply chains or consumer preferences, leading to imbalances between supply and demand.

ii. **Government interventions:**

As discussed above, government policies can intentionally create surpluses or shortages by influencing supply or demand.

iii. **Time lags:**

It takes time for producers to adjust production to changes in demand. This can lead to temporary surpluses or shortages.

iv. **Imperfect information:**

Consumers and producers may not have perfect information about market conditions, leading to imbalances in supply and demand.

Understanding how government interventions affect supply and demand, and why supply and demand may not always be equal, is crucial for analyzing market dynamics and predicting the consequences of policy changes. The price ceiling and price floor examples show that the quantity supplied does not necessarily equal the quantity demanded in a supply-and-demand model. The quantity supplied need not equal the quantity demanded because of the way we defined these two concepts. We defined the quantity supplied as the amount firms want to sell at a given price, holding other factors that affect supply, such as the price of inputs, constant.

The quantity demanded is the quantity that consumers want to buy at a given price, if other factors that affect demand are held constant. The quantity that firms want to sell and the quantity that consumers want to buy at a given price need not equal the actual quantity that is bought and sold.

When the government imposes a binding price ceiling of p on gasoline, the quantity demanded is greater than the quantity supplied. Despite the lack of equality between the quantity supplied and the quantity demanded, the supply-and-demand model is useful in analyzing this market because it predicts the excess demand that is actually observed.

We could have defined the quantity supplied and the quantity demanded so that they must be equal. If we were to define the quantity supplied as the amount firms actually sell at a given price and the quantity demanded as the amount consumers actually buy, supply must equal demand in all markets because the quantity demanded and the quantity supplied are defined to be the same quantity.

It is worth pointing out this distinction because many people, including politicians and newspaper reporters, are confused on this point. Someone insisting that "demand must equal supply" must be defining supply and demand as the actual quantities sold.

Because we define the quantities supplied and demanded in terms of people's wants and not actual quantities bought and sold, the statement that "supply equals demand" is a theory, not merely a definition. This theory says that the equilibrium price and quantity in a market are determined by the intersection of the supply curve and the demand curve if the government does not intervene. Further, we use the model to predict excess demand or excess supply when a government does control price. The observed gasoline shortages during periods when the U.S. government-controlled gasoline prices are consistent with this prediction.

**Multiple Choice Questions**

1. Which of the following is a policy that governments use to shift supply curves?

(a) Taxation

(b) Consumer preference changes

(c) Population growth

(d) Increase in disposable income

Answer: (a)

2. What is the primary effect of licensing laws on the labor market?

(a) They increase the number of workers in an occupation.

(b) They raise the average quality of the workforce.

(c) They decrease wages across the board.

(d) They shift the supply curve to the right.

Answer: (b)

3. A quota on imports is likely to result in which of the following market conditions?

(a) Increased consumer choice

(b) Lower prices for domestic goods

(c) Higher prices and reduced quantities of the imported goods

(d) Equal supply and demand for all goods

Answer: (c)

4. What happens when a binding price floor is set above the equilibrium wage?

(a) It leads to increased employment opportunities.

(b) It results in a surplus of labor.

(c) It has no effect on the labor market.

(d) It increases the quantity of labor demanded.

Answer: (b)

5. Which of the following statements accurately describes why supply and demand might not equal each other?

(a) Market dynamics are always stable and predictable.

(b) Government interventions can influence supply or demand.

(c) All consumers have complete information about the market.

(d) Supply and demand are always perfectly balanced.

Answer: (b)